



FLOOD MAPPING SERVICES

OPEN AND COMMERCIAL

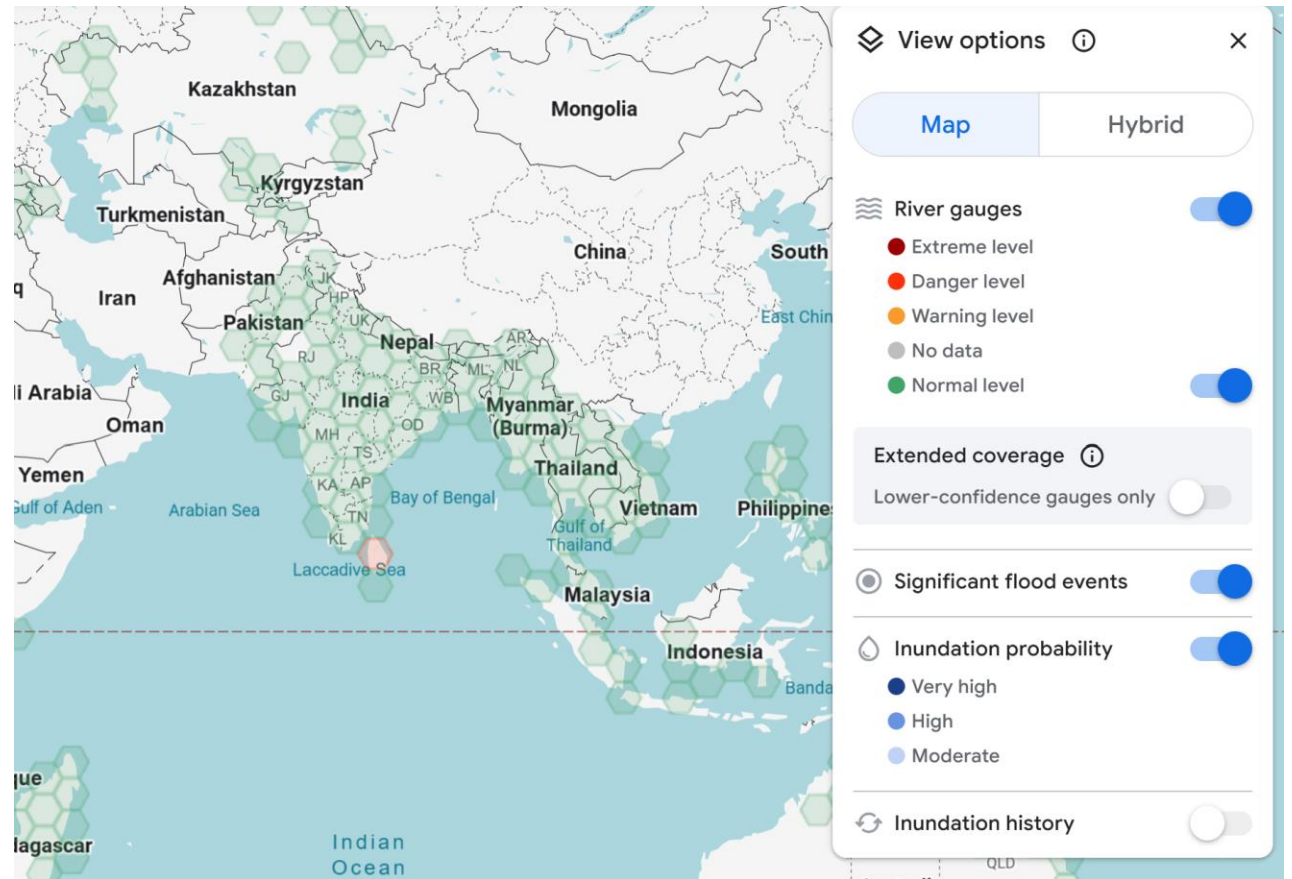
Aagya Dhungana

WHAT?

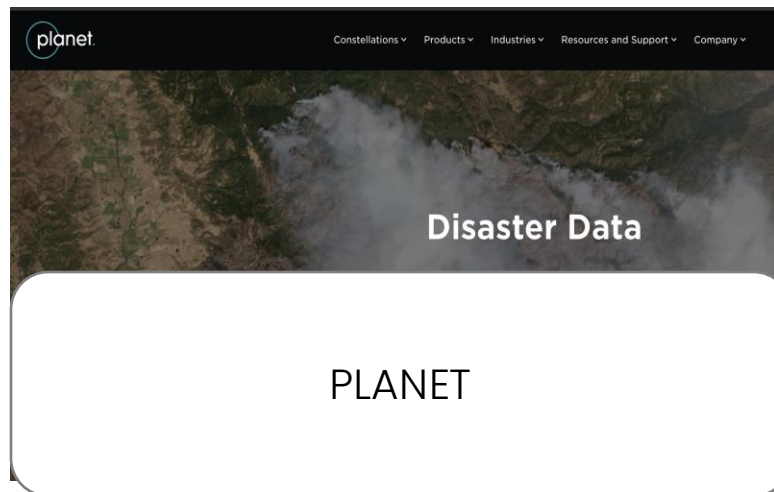
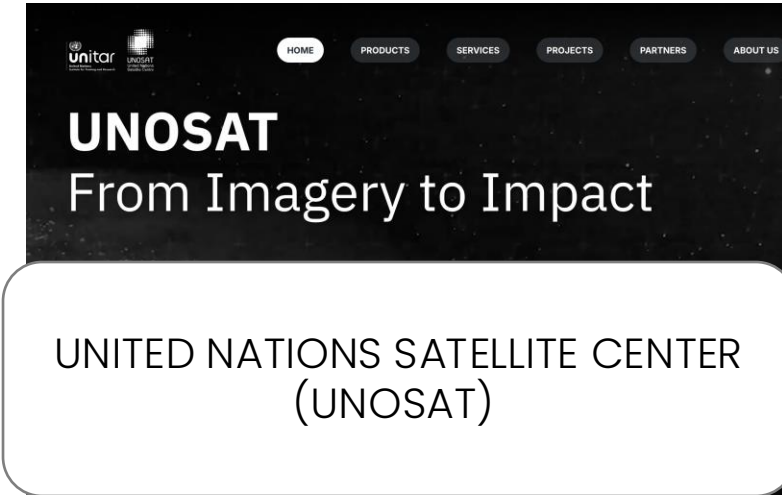
As the name suggests, services that help **detect**, **monitor**, and **predict** flooding using maps, satellite imagery, hydrological models, and ground data.

WHY?

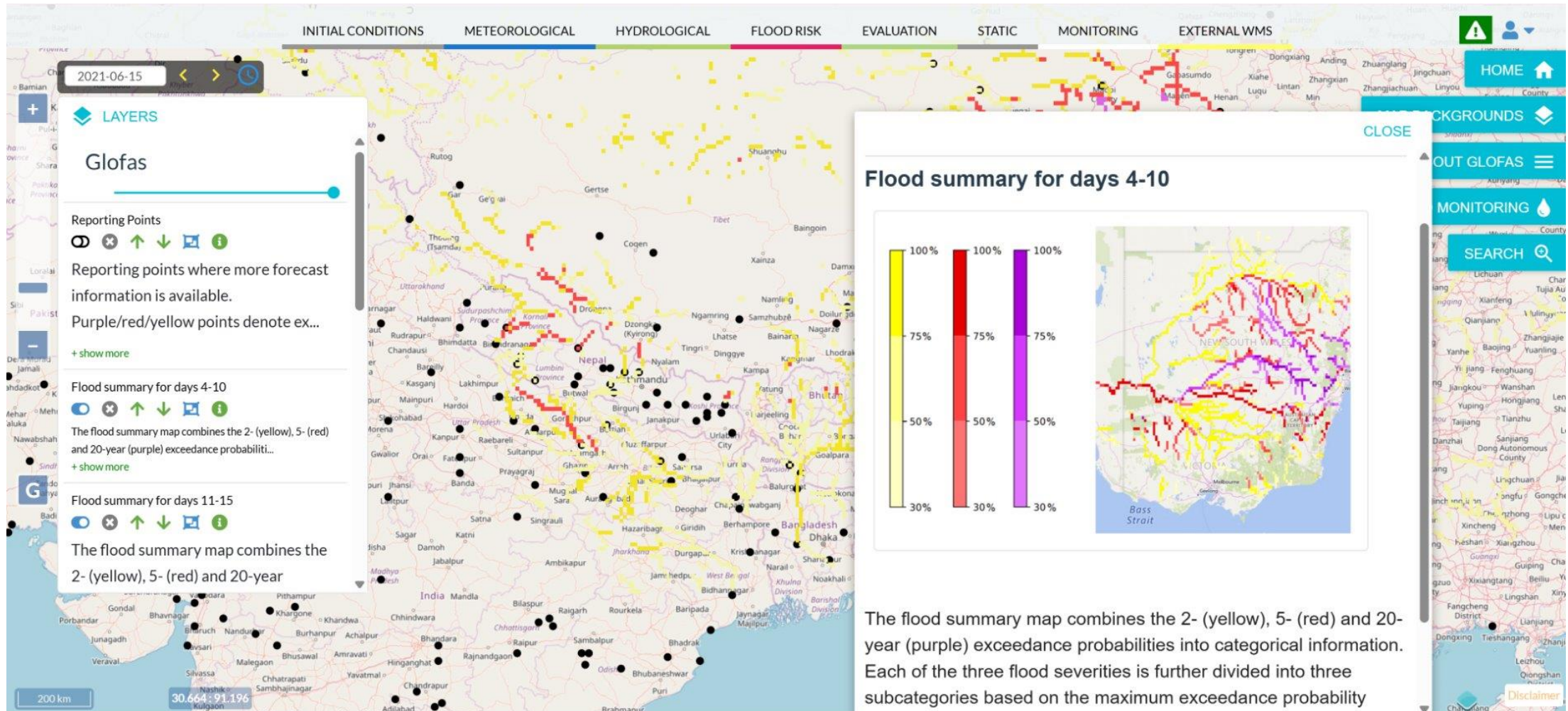
- Early Warning and Preparedness
- Rapid Disaster Response
- Damage Assessment & Recovery
- Urban Planning
- Evidence-Based Decision Making



FLOOD MAPPING SERVICES

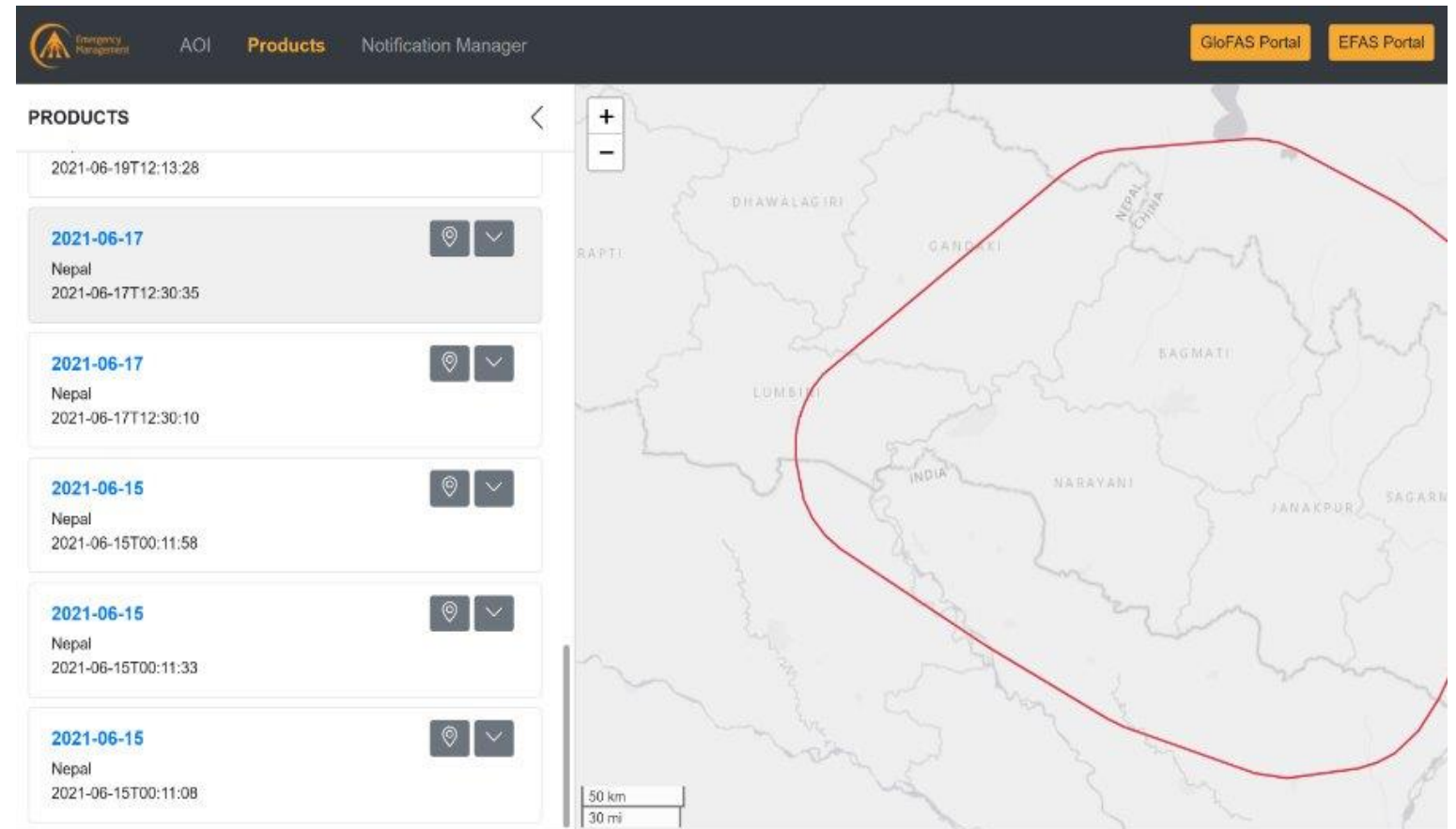


GLOFAS

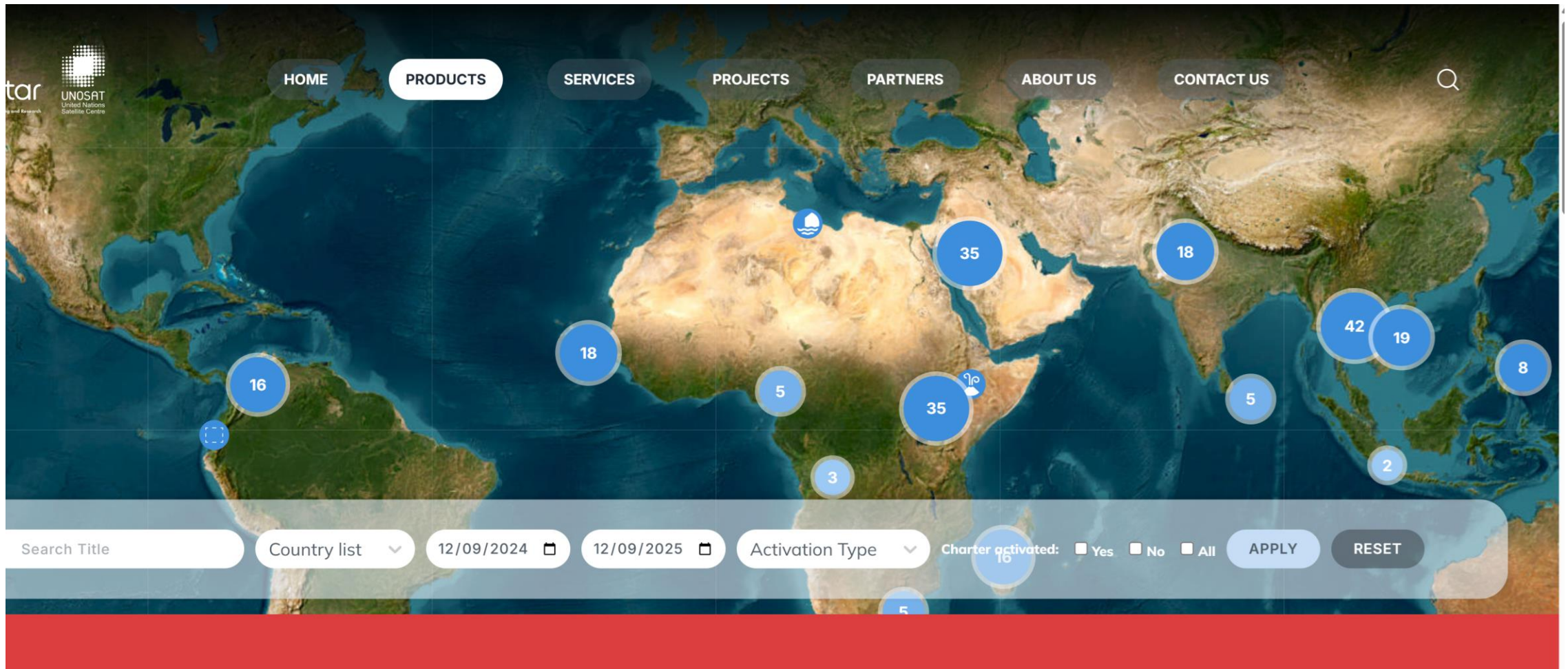


FEATURES OF GLOFAS

- Spatial Resolution: 5–9 km
- Temporal Resolution: Daily
- Operational Grid: 5 km
- Utilizes Copernicus Sentinel-1 SAR satellite data with in-situ gauging system

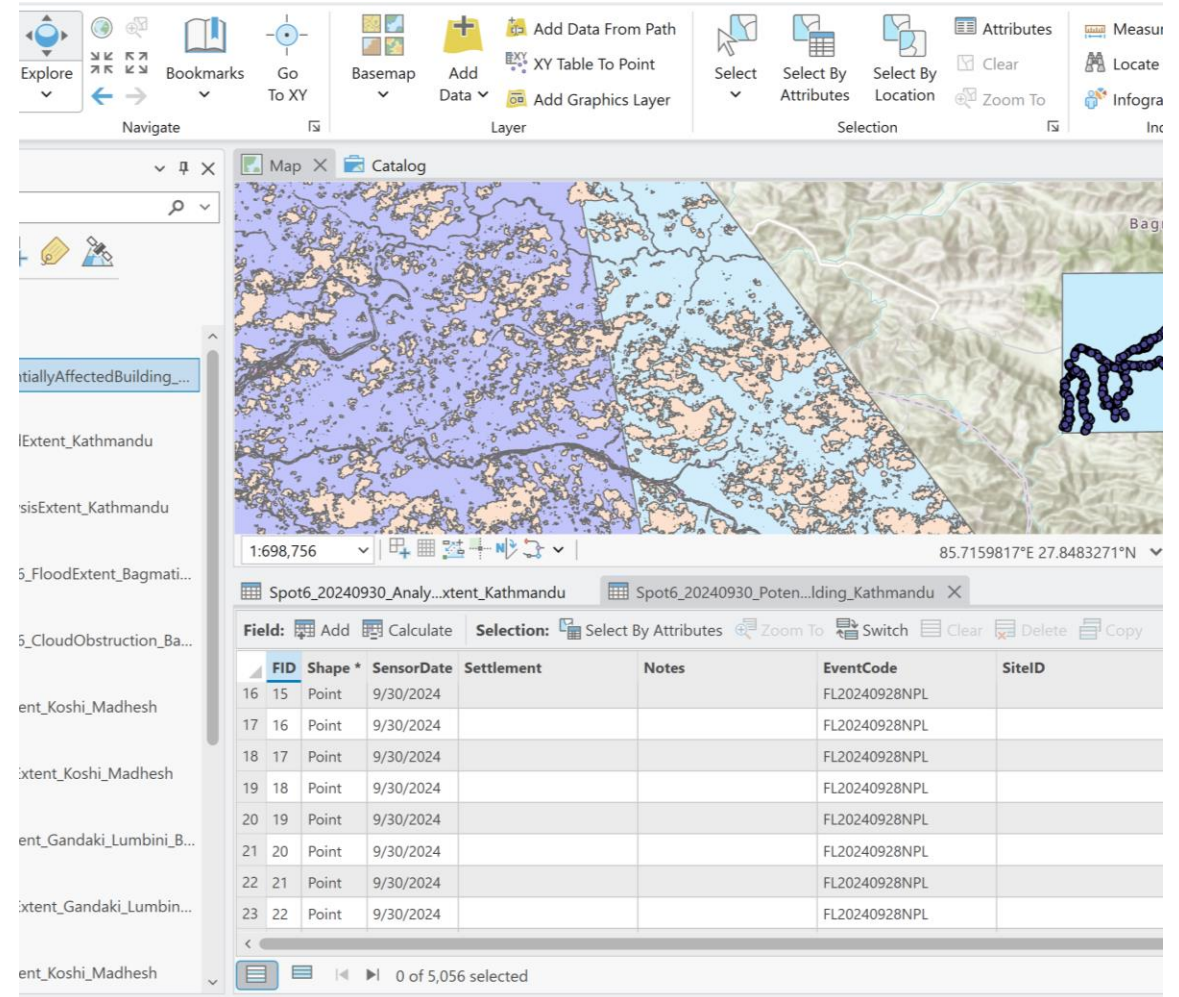


UNOSAT



FEATURES OF UNOSAT

- Provides analysis ready and analysed data (vector, raster or pdf)
- Dedicated for humanitarian response and recovery
- Analysis ready dashboards post event



PLANET

The screenshot displays the Planet Labs web application interface. On the left, a sidebar contains navigation icons and a 'Daily results' section. The 'Daily results' section lists three satellite scenes from December 2025, each with a thumbnail, date, 'PlanetScope Scene' label, resolution ('3m/px'), and coverage ('100%'). The first two scenes are from December 8 and 7, 2025, each with a '1 ITEM >' link. The third scene is from December 6, 2025, with a '2 ITEMS >' link. Below the list are settings for API and an 'ORDER (0)' button. The main area shows a satellite map of Melamchi River, Nepal, with a search bar at the top. A white rectangular box highlights a specific area on the river. The map includes labels for 'Indrawati river', 'Bhotang Road', 'होटल मेलम्ची रिभर साइड', 'होटल नेवा', 'होटल मेलम्ची साइड', and 'Melamchi hospital'. A timeline at the bottom of the map shows dates from 09 to 08, with 'Dec 2025' centered. The bottom status bar displays coordinates (27.84118° N, 85.56925° E), zoom level (15.33), resolution (3.36 m/px), and a scale bar (500 m). The footer includes copyright information: '©2022 Planet Labs PBC © Mapbox © OpenStreetMap'.

Daily results

FILTER DATES SAVE SEARCH

DESELECT ALL

December 8, 2025
PlanetScope Scene
3m/px 100%
1 ITEM >

December 7, 2025
PlanetScope Scene
3m/px 100%
1 ITEM >

December 6, 2025
PlanetScope Scene
3m/px 100%
2 ITEMS >

API {} ORDER (0)

Melamchi River, Nepal

Indrawati river

Bhotang Road

होटल मेलम्ची रिभर साइड

होटल नेवा

होटल मेलम्ची साइड

Melamchi hospital

09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 01 02 03 04 05 06 07 08
Dec 2025

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27.84118° N, 85.56925° E

15.33 3.36 m/px

500 m

FEATURES OF PLANET

- Pre-processed satellite imageries
- Spatial resolution 0.5 m
- Temporal resolution: Daily and on demand



Pelican



PlanetScope

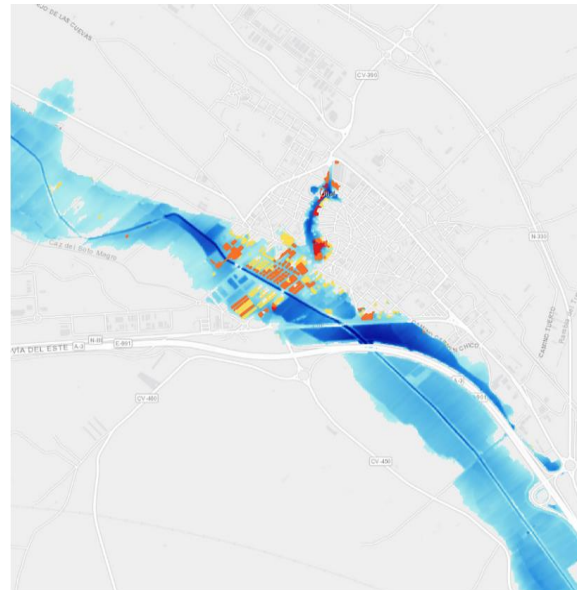


ICEYE



FEATURES OF ICEYE

- Spatial Resolution : up to 25–50 centimeter
- Global coverage daily
- Imageries: SAR
- 35 satellite in its constellations
- Proprietary flood forecast tools



Utiel, Spain

RELEASE 8 | 9TH NOVEMBER

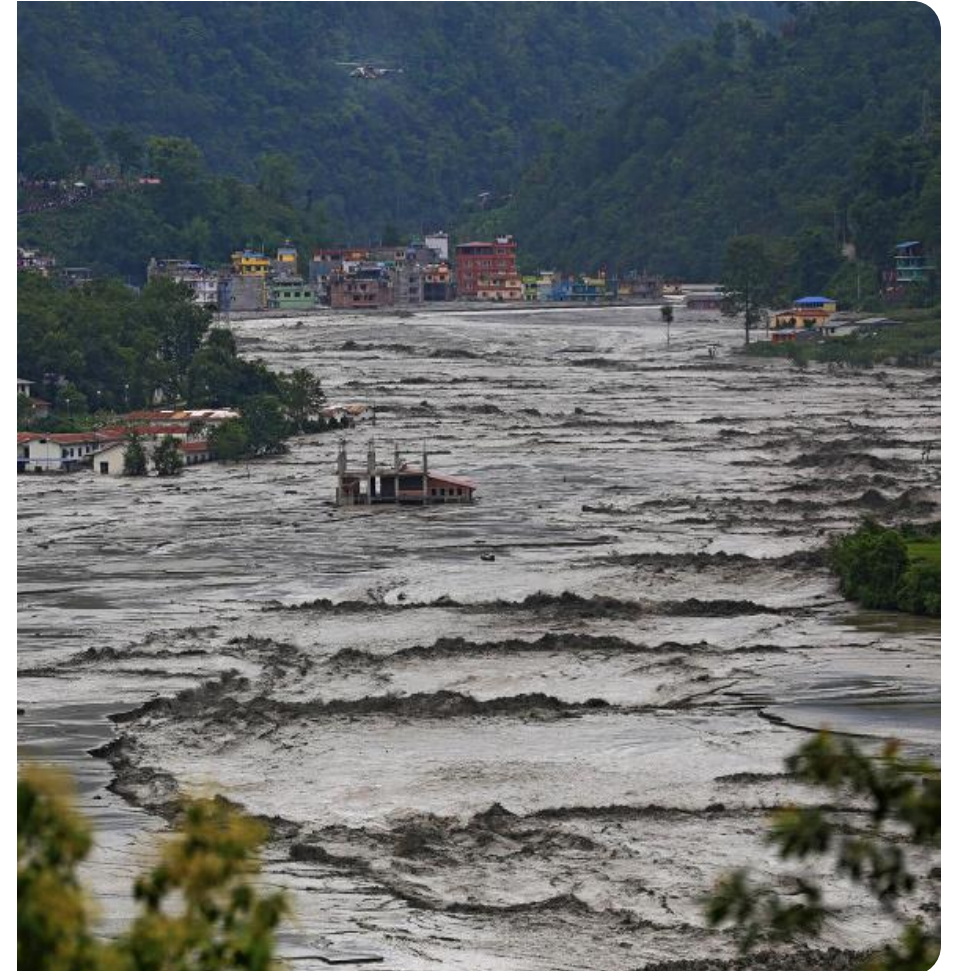
ICEYE analysis timeline

- **Monday, 28th October**
 - 06:19 GMT
Start of monitoring
- **Tuesday, 29th October**
 - 03:08 GMT
First SAR image acquisition
 - ~ 18:00 GMT
Storm peak
- **Wednesday, 30th October**
 - 17:33 GMT
First analysis release
- **Saturday, 9th November**
 - 19:42 GMT
Final analysis release

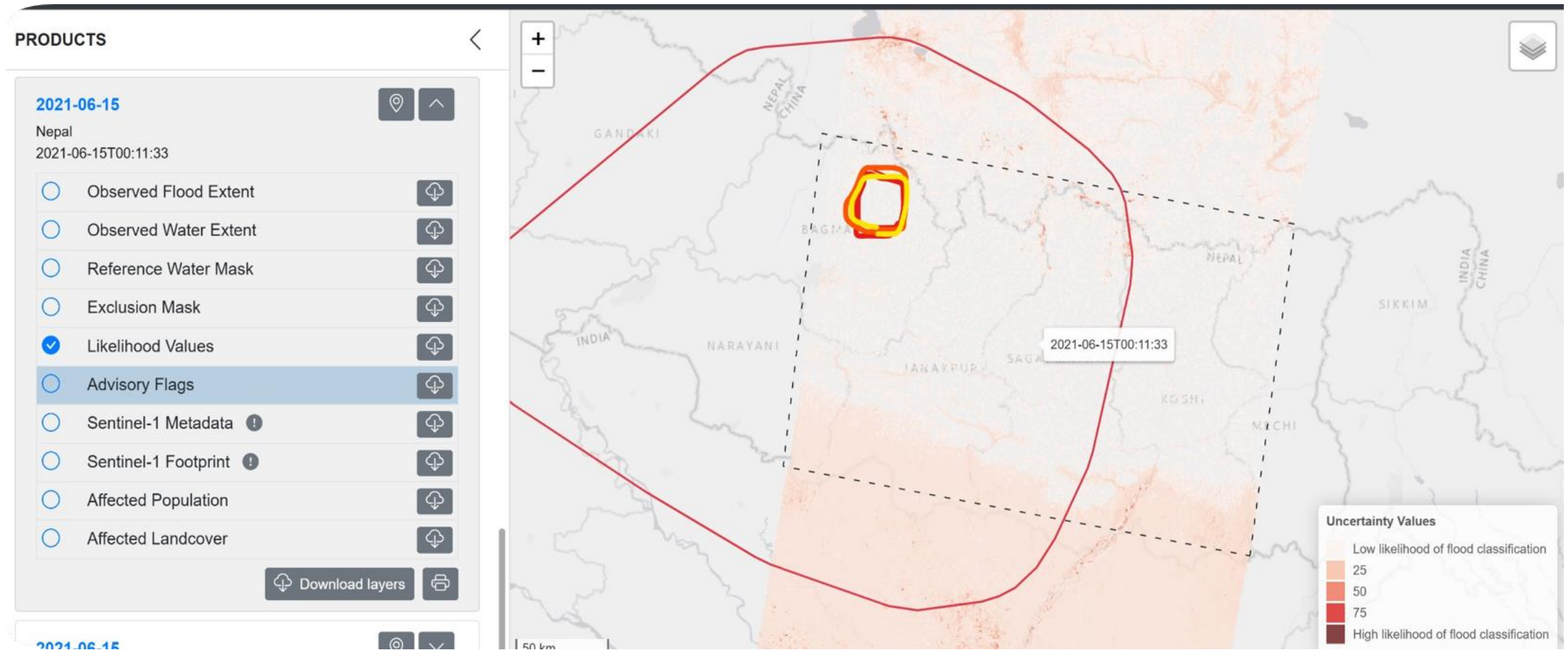
	GLOFAS	UNOSAT	PLANET	ICEYE
Primary Use Case	Early warning & forecasting & monitoring	Emergency rapid mapping (Post flood event)	Frequent change detection (commercial analytics)	Forecasting, near-real-time flood identification for insurance and governance
Key Products/ outputs	Ensemble discharge forecasts, downloadable datasets and APIs	Analyst-produced flood extents, impacts & damage maps (vector+raster+pdf+dashboards)	Daily optical scenes (3-5m), Skysat (50cm) imagery, APIs and AI analytics	SAR imagery, processed flood maps, Flood insights, on-demand reports
Pros	Free, Global Forecast, integrated with SAR imageries	Highquality analyst maps and vizualization for humanitarian focus.	Very high temporal revisit (daily), large archive, APIs for time-series analysis and change detection.	Rapid tasking, high spatial detail, operational flood depth/impact products for urgent response & insurance
Cons	Coarse hydrological forecast grid (~5 km), observed flood, footprints depend on SI coverage timing.	Not instantly available globally ; some services prioritized for UN / partners	Cloud/illumination-limited (bad in storms); tasking limited by availability and cost for high-res scenes.	Costly for continuous global coverage; requires contract and budget; pricing & availability can vary regionally.

CASE STUDY: MELAMCHI FLOOD 2021

- The Melamchi disaster was initiated by intense precipitation followed by GLOF and landslides preventing the flow.
- 17 human casualties, 23 reported missing
- 539 households damaged
- 4 days before the occurrence of the mass flow events, air temperature had risen significantly suggesting a transition to heavy snow melt.
- The devastating flood was the outcome of multifactor hazards which could have been predicted 2 days prior.

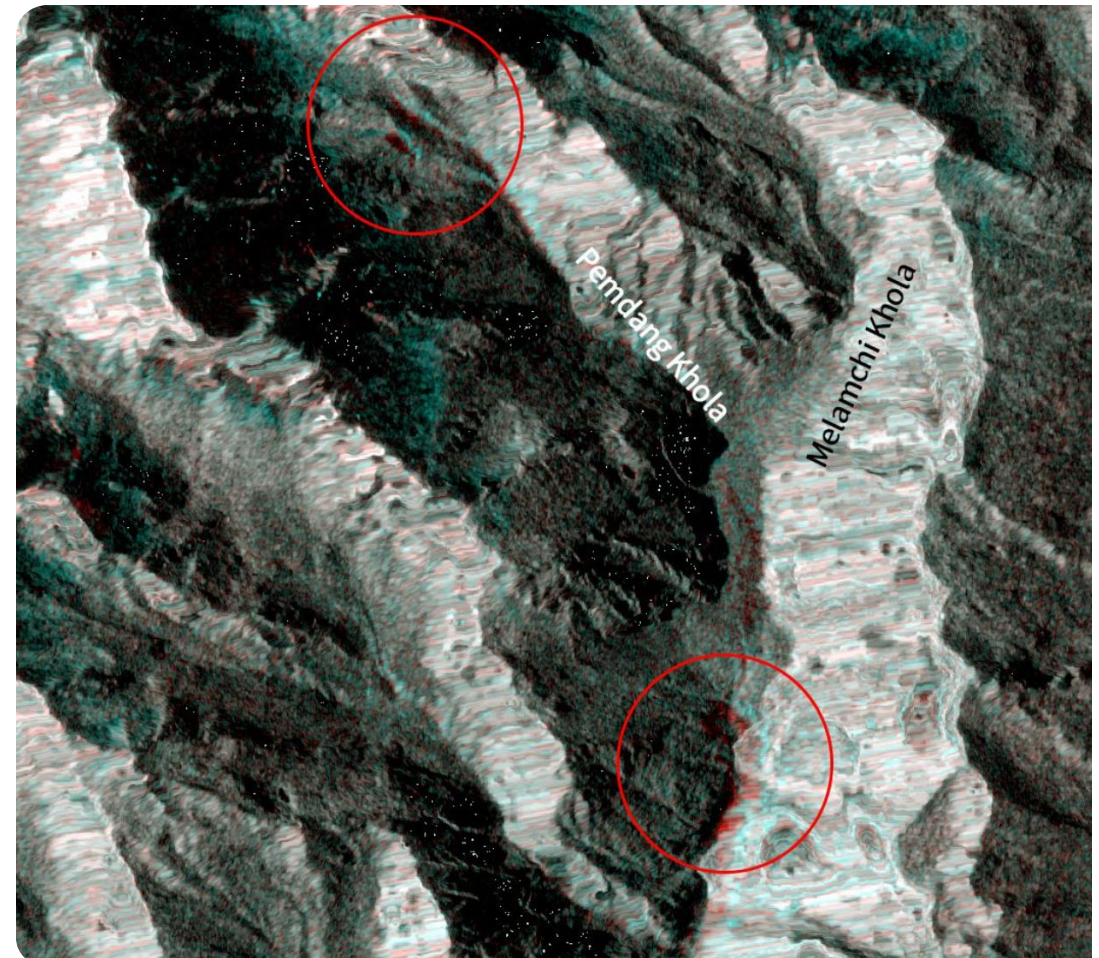


CASE STUDY: MELAMCHI FLOOD 2021

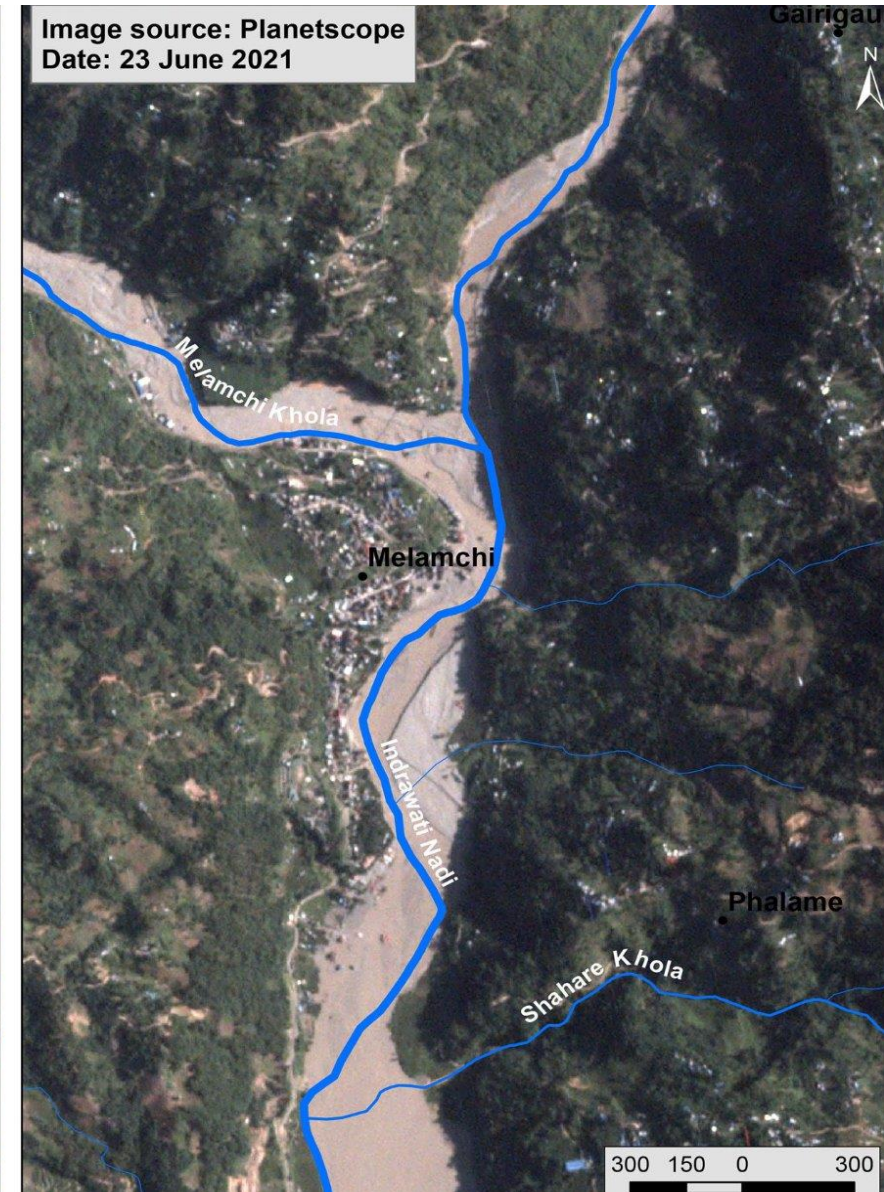
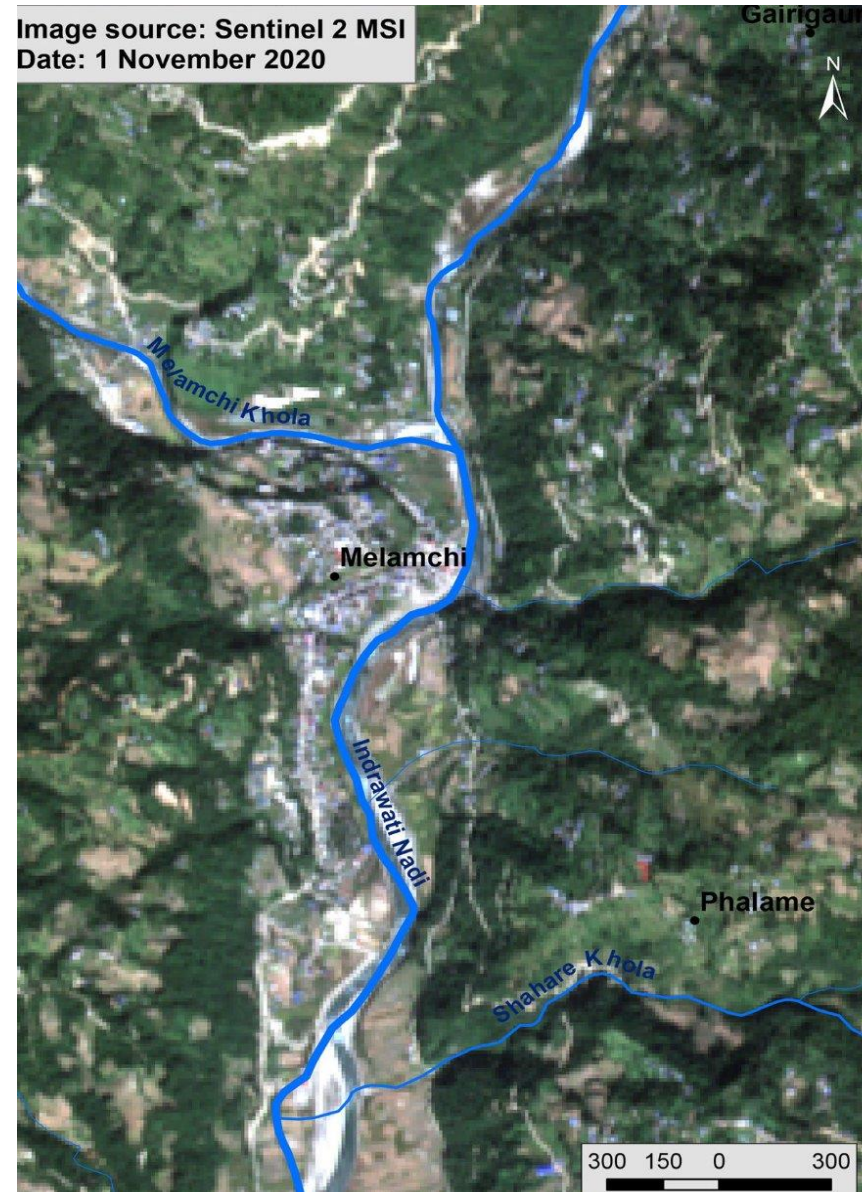


CASE STUDY: MELAMCHI FLOOD 2021

- The report on the flood event by ICIMOD utilized Sentinel-1 and Sentinel-2 image for the post disaster analysis.
- An understanding of the detailed breach mechanisms would only be possible through good quality high-resolution images and further field work.



WHEN TO USE WHAT



World Bank Document

CONCLUSION & RECOMMENDATIONS

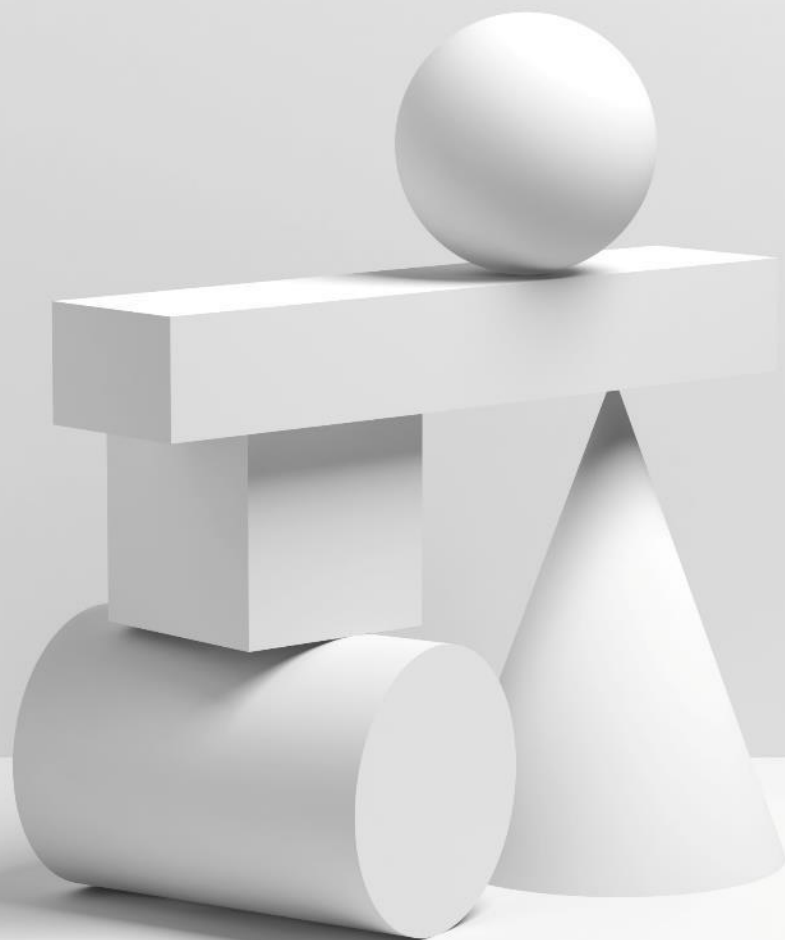
- SAR imageries and commercial high-resolution images provide detailed view required for Riverine floods and Glacial lake outburst floods.
- Open web mapping services like GLOFAS provides broad visualization of the flood event that supports global scale research
- Open flood map services provide a strong foundation and pool of analysis ready and hydrological data.
- For local level flood event, forecast and analysis, high resolution images obtained from commercial source backed by flood modelling applications have proven to provide good output.

REFERENCES

- Maharjan, S. B., Steiner, J. F., Shrestha, A. B., Maharjan, A., Nepal, S., Shrestha, M., Gupta, N. (2021). The Melamchi flood disaster: Cascading hazard and the need for multihazard risk management. International Centre for Integrated Mountain Development (ICIMOD).
- Dahal, Ranjan Kumar; Karelia, Hemang D.; Oommen, Thomas; Takamatsu, Masatsugu. *Melamchi Flood Disaster in Nepal: Damage and Risk Quantification with Drone Survey, Satellite-Based Land Displacement Analysis, and 2D Flood Modeling*. Washington, D.C. : World Bank Group.
<http://documents.worldbank.org/curated/en/099600006212237293>
- Adhikari, T. R., Baniya, B., Tang, Q., Talchabhadel, R., Gouli, M. R., Budhathoki, B. R., & Awasthi, R. P. (2023). Evaluation of post extreme floods in high mountain region: A case study of the Melamchi flood 2021 at the Koshi River Basin in Nepal. *Natural Hazards Research*, 3(3), 437–446.
<https://doi.org/10.1016/j.nhres.2023.07.001>

ADDITIONAL FLOOD MAPPING SERVICES

- <https://developers.google.com/flood-forecasting>
- <http://flood.umd.edu/>
- <https://www.fema.gov/flood-maps/know-your-risk>
- <https://www.floodbase.com/>
- <https://www.rsshhydro.lu/flood-risk-modeling-assessment>



QUESTIONS & DISCUSSION